Md Zesun Ahmed Mia

Specialist in ML for Yield Optimization & Process Control

Focused on leveraging data and AI to improve semiconductor fabrication processes. Combines a deep understanding of ML with practical knowledge of manufacturing workflows.

Education

08/2022–05/2027 (Expected) **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00*
Advisor: Dr. Abhronil Sengupta

08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00* | Thesis: Neuromorphic Computing for Lifelong Learning

02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, *CGPA: 3.81/4.00*

Industrial Experience

05/2025–07/2025 **Graduate Technical Intern**, *Intel Corporation*, Hillsboro, OR

- Developed AI-driven predictive models for process yield optimization and critical dimension (CD) control in semiconductor manufacturing, enabling automated recipe optimization.
- Implemented machine learning algorithms to predict optimal deposition parameters and automate recipe generation for high-volume semiconductor production.
- Designed comparative analysis framework between AI-based models and statistical models (JMP) for enhanced prediction accuracy and process window optimization.
- Conducted comprehensive data analysis using process characterization data to validate model performance and establish robust process control limits.

09/2020–07/2021 **R&D Engineer**, *SEMWAVES Ltd.*, London, UK, part-time

- Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
- Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

Technical Skills

ML for Semiconductors	Yield prediction, defect detection, process monitoring, EDA automation, ML/EDA integration, process-aware modeling, reliability analysis, AI accelerator development, HPC applications, parallel computing architectures
ML Tools	ONNX, TensorRT, PyTorch, TensorFlow, scikit-learn, Keras, Data Visualization (Matplotlib, Seaborn, Plotly), Pandas, NumPy, JMP, Jupyter, LaTeX
Hardware, EDA & Sim.	PCB design, Cadence Virtuoso, Spectre, HSPICE, TCAD, COMSOL, ModelSim, Synopsys (Design Compiler, PrimeTime, VCS), Oscilloscope, Signal Generator, CIM (Compute-In-Memory) Systems, Systolic Arrays, PIM, near-memory computing, memory-centric computing, data flow optimization
Programming	Python, CUDA, MATLAB, Verilog, Shell, C/C++, Bash, Linux/Unix
DTCO & STCO	Design-technology and system-technology co-optimization, cross-layer PPA, process-aware design, system-architecture codesign
Fabrication	Lithography, Etching, Deposition (CVD, HDP-CVD, PVD), E-beam, SEM, AFM, DSIMS, FTIR
AI Dev.	Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development
Collaboration	Git, Microsoft Office, Google Workspace

Academic Research and Teaching Experience

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08/2022– Present **Graduate Research Assistant**, *Penn State*, University Park, PA
- Developed compact models of FeFETs in Verilog-A and MATLAB from characterization data, enabling their integration into standard EDA toolchains for circuit design and verification.
- Designed an ML-based workflow to assess the system-level impact of fabricated spintronic device variations, enabling faster feedback for process optimization and EDA automation.

08/2024– 05/2025 **Graduate Teaching Assistant**, *Penn State*, State College, PA
- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
- Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.

02/2021– 08/2022 **Lecturer**, *University of Liberal Arts Bangladesh (ULAB)*, Dhaka, Bangladesh
- Taught undergraduate courses: Solid State Devices, Digital Circuit Design, Semiconductor Device Physics, Power Electronics.
- Developed lab modules and supervised projects on semiconductor devices and circuits.

Projects

Astromorphic Transformer — Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture leveraging neuron-astrocyte interactions to emulate self-attention mechanisms. Incorporated Hebbian and presynaptic plasticities with non-linearities and feedback, achieving superior accuracy and faster convergence on sentiment classification (IMDB), image classification (CIFAR-10), and language modeling (WikiText-2) tasks. See publication: [IEEE TCDS 2025].

Spintronic Device-Based Memory — Lead Student Researcher, 2023–Present. Led end-to-end fabrication of spintronic memory arrays using e-beam lithography, sputtering deposition, and lift-off patterning. Performed comprehensive electrical characterization (I-V, R-H loops, retention, endurance) and extracted device parameters for compact model development. Advanced non-volatile memory technologies for neuromorphic computing and in-memory processing applications.

Neuromorphic Cybersecurity with Lifelong Learning — Lead Student Researcher, 2023–2025. Developed a Hierarchical Dynamic Spiking Neural Network (D-SNN) for Network Intrusion Detection Systems combining static SNN detection (94.3% accuracy) with adaptive dynamic SNN classification using GWR-inspired structural plasticity and novel Adaptive STDP (Ad-STDP) learning. Achieved 85.3% overall accuracy on UNSW-NB15 benchmark in semi-supervised lifelong learning, demonstrating superior adaptation to new attack types while mitigating catastrophic forgetting (5.3% improvement over baseline). See publication: [arXiv], [ICONS 2025].

Select Publications

[1] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. "RMAAT: Astrocyte-Inspired Memory Compression and Replay for Efficient Long-Context Transformers". In: *International Conference on Learning Representations (ICLR)*. 2026. URL: `https://openreview.net/forum?id=sTkJdbVxsI`.

[2] **Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. "Trilinear Compute-in-Memory Architecture for Energy-Efficient Transformer Acceleration". In: *arXiv preprint arXiv:2604.07628* (2026).

[3] **Md Zesun Ahmed Mia**, Malyaban Bal, Sen Lu, George M. Nishibuchi, Suhas Chelian, Srini Vasan, and Abhronil Sengupta. "Neuromorphic Cybersecurity with Semi-Supervised Lifelong Learning". In: *Proceedings of the International Conference on Neuromorphic Systems (ICONS '25)*. IEEE Press, 2025, pp. 235–238. DOI: `10.1109/ICONS69015.2025.00043`.

[4] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. "Delving deeper into astromorphic transformers". In: *IEEE Transactions on Cognitive and Developmental Systems* (2025).

Recognitions

- Melvin P. Bloom Memorial Outstanding Doctoral Research Award (2026)
- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024),Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, IEEE TNNLS (2025), Design Automation Conference (DAC) (2025), IEEE MWSCAS (2025)
- Student Member, IEEE (2015-Present), Executive Member, EDS, IEEE BD (2021-2022)

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